

Performance Analysis in the Brazilian Soccer League: Applying Machine Learning Techniques to Team Evaluation

José Vinicius Boaventura Barbeiro¹, Bruno Samways dos Santos²

¹ Universidade Tecnológica Federal do Paraná (UTFPR), Londrina, Brazil

² Advisor. Universidade Tecnológica Federal do Paraná (UTFPR), Londrina, Brazil

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Abstract

This study presents a method for analyzing the sports performance of teams through artificial intelligence, using data from the Série A of the Brazilian Championship. Machine learning methods were applied, highlighting K-Means clustering for identifying patterns among teams, Random Forest for pointing out the most relevant variables, and SHapley Additive exPlanations (SHAP) analysis to interpret the importance of these variables in each profile. The clustering revealed six distinct groups of teams, ranging from aged and less competitive squads to young, offensive, and disciplined teams. Variables such as the number of goals, average age of the starters, betting market odds, and number of fouls stood out as determinants for performance. The results demonstrate how the application of artificial intelligence enhances game interpretation, offering deeper insights and more solid bases for strategic decisions, contributing to understanding the factors influencing team performance, and strengthening sports analysis based on objective data and advanced methods.

Keywords: Machine Learning; Clustering; Explainable Artificial Intelligence; Soccer

1 Introduction

Artificial Intelligence (AI) has been revolutionizing the world's most popular sport: Soccer. As a global industry that generates billions of dollars annually (Fimmanò, 2024), soccer is increasingly reliant on advanced technologies to optimize performance, engage fans, and achieve results. In this context, AI enables large-scale, precise, and personalized analyses, transforming how clubs, athletes, and governing bodies manage, play, and commercialize the sport. Through machine learning, it becomes possible to predict match outcomes with greater accuracy, analyze tactics in real time, and assess both individual and team performance. As a sporting and economic phenomenon, soccer is entering a new era driven by the power of data and artificial intelligence (Andreff, 2021).

Due to its democratic and universal nature, soccer goes beyond being merely a sport. In recent years, however, the game has also solidified its status as a major industry, where success on the field is increasingly tied to investments in infrastructure, technology, and performance analysis. The modernization of club management, especially with the adoption of the *Sociedade Anônima do Futebol* (SAF) model, or Soccer Joint-Stock Companies, has contributed to greater professionalization, enabling teams to incorporate advanced tools for data analytics and tactical scouting.

This evolution has enabled a more scientific approach to the game, incorporating statistical techniques and machine learning to identify patterns in team performance. The relationship between financial resources and sporting success had already been evident for some years. However, with the growing availability of data and the development of more sophisticated analytical methods, it has now become possible to extract tactical insights with an unprecedented level of precision (Okazaki et al., 2012). This progress allows for a deeper understanding of game strategies and how investments can be optimized to improve on-field results (Rein and Memmert, 2016).

In this context of the increasing transformation of sports information into digital data, quantitative performance analysis has emerged as a fundamental strategic tool, reshaping how teams are evaluated and developed tactically. The use of AI in sports databases enables the identification of tactical and behavioral patterns with a high degree of accuracy (Pisaniello, 2024).

Through machine learning algorithms, it is possible to analyze large volumes of performance data, such as positional metrics, offensive transitions, and defensive effectiveness. These tools provide quantitative support for the evaluation of strategies, optimizing decision-making processes, and allowing for evidence-based tactical adjustments. This approach represents a significant advancement in sports analysis, integrating technological and scientific methods into team preparation (Pisaniello, 2024).

Considering this context, the present study aimed to analyze how machine learning techniques, particularly clustering methods combined with classifiers (Random Forest) and feature importance analysis with SHapley Additive exPlanations (SHAP), can characterize team performance in the Brazilian Championship. The research specifically focuses on identifying collective performance patterns and conducting comparative analyses between teams using competition performance data. This approach not only enhances the understanding of the league but also provides a foundation for more effective strategies in analysis and decision-making. In addition to this introductory section, the remainder of the article is organized into four main sections: Section 2 discusses previous studies that apply machine learning in soccer; Section 3 describes the data, methods, and tools used; Section 4 presents the results of the clustering, Random Forest, and SHAP analysis, including charts, descriptive analyses, and practical implications; finally, Section 5 outlines the conclusions and contributions of the study. The references are provided at the end of the article.

2 Correlated Works

The use of machine learning in sports analysis has been explored by many authors in the past years with many applications and different areas. These studies are focused on predicting match results, finding playing styles, game strategies, or identifying features that can help teams better understand how they can prepare for opponents. Soccer, being one of the most popular sports in the world, has certainly gained space in this arena. Analyzing some studies related specifically to soccer we can find many of them that use clusterization algorithms to find how matches, players or teams are related to each other.

Diquigiovanni and Scarpa (2018) developed an innovative hierarchical clustering method that enables the characterization of playing styles within Italian Serie A teams. This method identified the relationship between playing styles and performance during the 2015–2016 season, focusing on the effect of playing styles on the number of goals scored. They were able to prove the key role of one of them in the match results. Using a different technique, Fernandez-Navarro et al. (2016) also identified patterns and playing styles in teams playing the Spanish La Liga and the English Premier League. Their work utilizes Principal Component Analysis (PCA) and distributes the teams into clusters based on this factor analysis, allowing them to create a playing style profile to help teams better prepare for opponents in competition.

Bilek and Ulas (2019) investigated how situational variables and performance indicators such as fouls, passes, offsides, yellow and red cards, among others, could affect a match outcome considering the quality of the opposition. Their results confirmed the importance of some of these variables and the influence of the opposite team on match results.

Other variables were investigated by Lago (2009), such as match location and possession strategies to better understand their influence on the match results and how teams can alter their playing styles during the match in response to match status. The author examined twenty-seven games from the 2005–2006 domestic league season in a professional Spanish soccer team. The findings emphasize the relevance of accounting for these variables when studying tactics and performance in soccer. Malamatinos (2022) conducted a study using multiple Machine Learning methods to predict the outcome of games in the Greek Super League. The results found the most important features that affect the results as Home Team Form and Away Team Form, for most models. The Greek Super League was found to be the most predictable among the 3 leagues analyzed, reporting 67.73% of accuracy with CatBoost.

Musa et al. (2021) analyzed the Asian beach soccer tournament through hierarchical agglomerative cluster analysis, identifying relevant essential performance parameters that could describe winning and losing performance in the tournament. A similar analysis was conducted to find predictive variables of winning and losing in the Belgian soccer division by Geurkink et al. (2021). Their results using Extreme Gradient Boosting showed an accuracy of 89.6% and using TreeExplainer they found the most important variable for the prediction to be “total shots

on target from attacking penalty box”, followed by player’s physical indicators as number of accelerations, runs and distance, and contextual variables as Home/Away match.

3 Materials and Methods

This research employs both unsupervised machine learning methods (for the clustering task) and supervised methods (for selecting feature importance), in order to analyze and assess the influence of different variables on the performance of teams in the Série A of the Brazilian Championship. To ensure the robustness of the study, the data were consolidated from two complementary sources.

The first data source used comprises match statistics from the Brazilian Football Championship (Série A) spanning the years 2003 to 2024. The records were obtained through the platform basedosdados.org, which consolidates information originally sourced from Transfermarkt, a football-specialized database. This dataset includes game metrics and team characteristics for each match. The data are structured separately for home and away teams.

To complement the analysis, a second dataset was incorporated, obtained from the portal football-data.co.uk, containing probabilistic information on matches from the Brazilian Championship between 2012 and 2024. These data represent the average odds offered by betting companies for three possible outcomes: home win, away win, and draw. The inclusion of these probabilistic variables enriches the study by adding a market perspective and mathematical expectations regarding match outcomes.

The integration of the datasets enabled a more comprehensive analysis by combining descriptive statistics with implied probabilities, thereby expanding the scope of the investigation. After consolidating the data, the year 2019 was specifically selected for modeling, as it presented the most complete set of information across both datasets.

For the clustering task, the K-Means algorithm was adopted, a method particularly suitable given the predominantly numerical nature of the variables and the objective of identifying multivariate performance patterns among teams. K-Means is simple and flexible (Nayini et al., 2017), and while easy to implement, it is also highly efficient and widely used across various

fields of knowledge (Chong, 2021).

The original data (organized by matches) were transformed into a team-level structure, generating the variables presented in Table 1, which include totals, means, standard deviations, minimums, and maximums of the defined metrics.

Table 1: Variables used in the team-level dataset.

Variable	Variable	Variable
Total Goals First Half	Total Goals	Total Offsides
Average Odds	Total Shots Off Target	Standard Deviation of Shots
Total Shots	Average Shots	Standard Deviation of Offsides
Minimum Shots Off Target	Maximum Shots Off Target	Total Set Pieces
Minimum Set Pieces	Maximum Set Pieces	Total Fouls
Minimum Fouls	Maximum Fouls	Average Age
Minimum Age	Maximum Age	Victories Playing Against Team A
Victories Playing Against Team Bellow	Matches Against Team Above	

To neutralize the impact of differing variable scales given the distance-based nature of the K-Means algorithm, the dataset was normalized using the MinMaxScaler method, converting all values to the $[0, 1]$ range. This standardization ensured that metrics with different units (such as average age and number of goals) had equivalent weight in the cluster analysis. As a preliminary step to clustering, a Principal Component Analysis (PCA), proposed by Hotelling (1933), was conducted using five components that captured approximately 75% of the total data variance. This dimensionality reduction served two key purposes: (1) it eliminated redundancies among highly correlated variables, particularly relevant in interrelated sports metrics. This transformation is based on the decomposition of the covariance matrix into eigenvalues and eigenvectors, where the eigenvectors determine the directions of the new variables and the eigenvalues indicate the variance explained by each component; (2) it preserved the essential structure of the data by retaining most of the original information (Jolliffe and Cadima, 2016).

To determine the optimal cluster configuration, the silhouette score was employed. This metric revealed that the configuration with six clusters yielded the highest average silhouette score, indicating a more coherent clustering structure compared to solutions with fewer clusters. The choice of six clusters is justified not only by the statistical criterion but also by its ability to preserve strategic heterogeneity among teams, avoiding the excessive generalization observed in broader groupings. This refined segmentation enabled the identification of distinct tactical

patterns and variations in operational efficiency.

The K-Means algorithm, selected for this analysis, stands out for its effectiveness in segmenting numerical data into cohesive groups, operating through an iterative process that minimizes the intra-cluster sum of squares. Specifically, the method: (1) initializes k centroids randomly; (2) assigns each observation to the nearest centroid, calculated using Euclidean distance; and (3) recalculates the centroids as the mean of the observations in each cluster, repeating this process until convergence (Bramer, 2016). This approach ensures the progressive minimization of intra-cluster variance and is particularly suitable for our preprocessed sports dataset, where the goal is to maximize tactical homogeneity within each group while maintaining clear strategic distinctions between clusters (Bezdek, 2022).

Complementing the cluster analysis, a Random Forest classifier was employed to assess the importance of the variables in the formation of the clusters. This method quantifies the contribution of each feature through the mean decrease in accuracy when its values are permuted, enabling the identification of the most influential attributes for the resulting segmentation (Géron, 2019). To further enhance interpretability, the SHAP (SHapley Additive exPlanations) technique was additionally applied. Based on cooperative game theory, SHAP assigns fair values to each variable's contribution to a model's prediction. It sheds light on the underlying reasoning behind machine learning model outputs (Lan et al., 2024), thus overcoming the limitations of black-box models that typically offer only prediction results after training (Ferreira et al., 2024). This approach allowed us to: (1) identify the most discriminative variables across clusters using Random Forest; (2) understand the direction and magnitude of each feature's impact on specific clusters through SHAP values, which capture how the presence or absence of a variable alters the clustering prediction. Furthermore, the decomposition into Shapley values made it possible to uncover variable interactions, revealing how specific combinations of attributes influence the distinct tactical profiles of the six identified clusters.

Figure 1 illustrates the complete process: from data integration and transformation (with PCA and normalization), through clustering using K-Means ($k = 6$ clusters), to the interpretive analysis via Random Forest and SHAP, identifying the most relevant variables for the detected performance-based groupings.

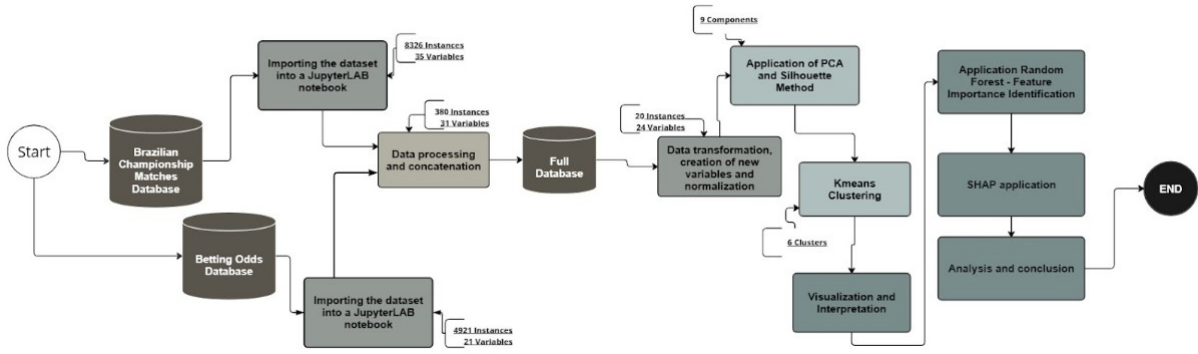


Figure 1: Flowchart of the analysis process for the Brazilian Championship dataset.

4 Results and Discussion

4.1 Distribution of Teams by Cluster

Figure 2 represents the distribution of teams in their respective cluster for better understanding in the following analyses, while Figure 3 shows the top 5 relevant features concerning the clusters created.

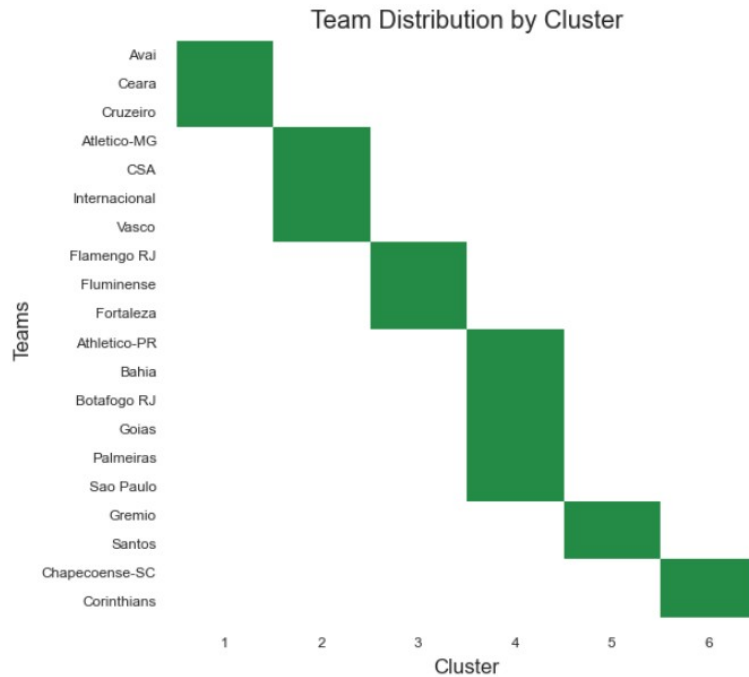


Figure 2: Heatmap distribution by cluster.

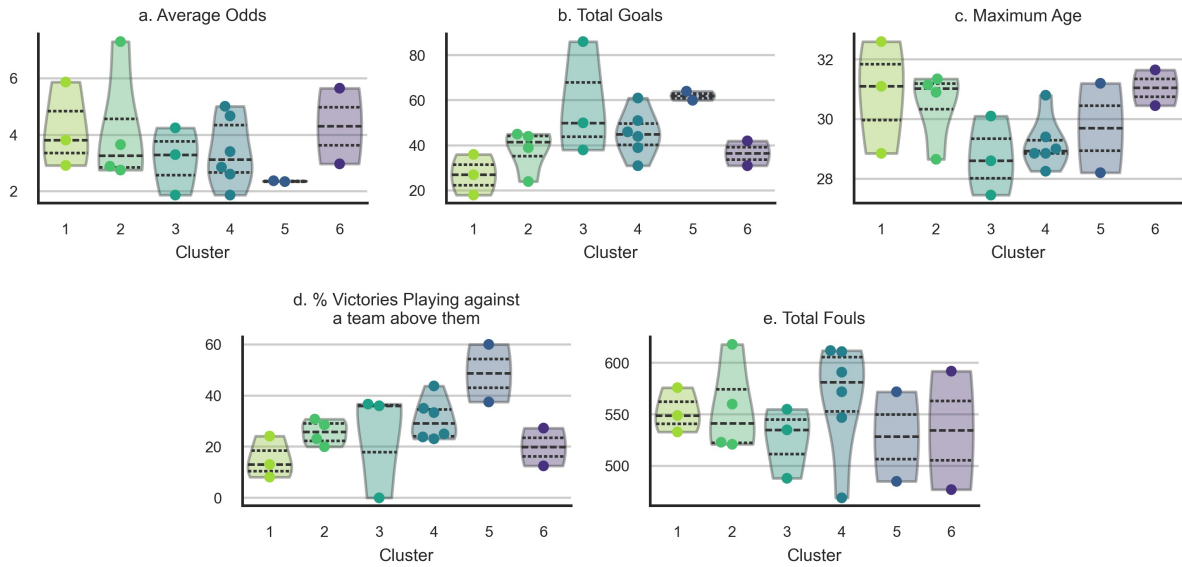


Figure 3: Graph Top 5 features importance x Cluster distribution.

4.2 Top 5 Features Importance x Cluster Distribution

Regarding Figure 3(a), the variable *Average Odds* reflects the betting market's perception of each team's average probability of winning, where higher odds values indicate a lower probability of victory. Notably, Clusters 1 and 6 exhibit the highest odds in the championship. This is consistent, as these clusters include Avaí, Ceará, and Chapecoense—teams that demonstrated low offensive performance and limited expectations of winning, both at home and away. These teams finished at the bottom of the league table, with Avaí and Chapecoense being relegated to the second division. Cluster 3 is particularly significant for its consistently low odds, indicating a high probability of victory. This can be attributed to the dominant performance of the teams in this cluster throughout the season, especially Flamengo. Cluster 5 displays the lowest average odds among all clusters and includes Grêmio and Santos, which maintained relatively consistent performances during the championship.

Analyzing Figure 3(b), the *Total Goals* variable represents the total number of goals scored by each team during the championship. Clusters 1 and 2 recorded the fewest goals in the league, characterized by unproductive attacks with limited creativity and offensive presence. Cluster 3 includes the team with the highest number of goals, Flamengo. Clusters 4 and 5 are composed of teams with offensive potential but display varied playing styles, as some teams scored many goals while others scored relatively few. Cluster 6 consists of teams with low

offensive output, but which often secured victories by narrow scorelines.

Regarding Figure 3(c), this variable represents the team with the highest average age among starting players. Two standout clusters in this chart are Cluster 1, which consists of lineups with older players meaning more experienced squads, though sometimes lacking physical intensity, which can be detrimental on the field, and Cluster 3, composed of teams with the youngest squads among all clusters, typically showing higher offensive intensity when analyzed in relation to other variables.

As for Figure 3(d), the variable represents the percentage of victories against higher-ranked teams. Clusters 1 and 2 are composed of teams that struggle to overcome opponents with stronger squads. In contrast, Clusters 5 and 6 include teams that perform well against superior opponents, often traditional clubs that, despite being in a poor phase, still manage to remain competitive. Teams such as Corinthians, Santos, and Palmeiras are present in these clusters.

Still discussing the findings from Figure 3, an interesting correlation was observed between total goals and fouls: teams with low offensive output tend to commit more fouls, often as a way to compensate for their lack of creativity and control of the game by adopting a more physical and defensive style of play (Silva et al., 2009). Furthermore, teams with older squads generally display a slower pace, which contributes to a higher number of fouls committed, as lower intensity makes it harder to recover defensively and leads to more interruptions in play to stop the opponent (Neto et al., 2020).

4.3 SHAP Analysis

To verify deeply the most important features in each cluster formation, Figure 4 illustrates the SHAP values for each cluster.

Figure 4(a) illustrates Cluster 1, which groups teams with older squads on average (Poli et al., 2018), including even among their younger players. These teams tend to struggle offensively and show low competitiveness against opponents ranked above them. The presence of Avaí, which was relegated at the end of the season, and Ceará, which also fought against relegation, reinforces that this cluster represents teams with weak overall performance.

As shown in Figure 4(b), Cluster 2 includes younger teams that make little use of set



Figure 4: Graph SHAP Analysis.

pieces. There is no clear trend when it comes to their performance against higher-ranked opponents, but these teams tend to take good advantage of playing at home, often securing wins against weaker sides. Teams such as CSA and Cruzeiro, both relegated in 2019, along with Atlético-MG, Internacional, and Vasco form a group with highly heterogeneous results, yet they share similar tactical and structural profiles.

The analysis of Figure 4(c) reveals that Cluster 3 is characterized by teams with young and dynamic squads, typically top-table teams that, in most matches, faced lower-ranked opponents and were favored by the odds. These teams tend to have reliable defenses and likely adhere to fair play principles, as indicated by a low number of fouls committed (Bunker and Susnjak, 2022). They employ offensive and efficient playing styles, with little reliance on set pieces, favoring either direct attacks or well-constructed plays combined with strong tactical control. The presence of Flamengo, the league champion, and Fortaleza, a well-structured team, reinforces this pattern of dominant and consistent performance throughout the 2019 season.

Regarding Figure 4(d), Cluster 4 is represented by experienced teams, with both high

average and maximum player ages. These teams are defensively disciplined and register a lower number of set pieces, which may suggest a playing style based on physical pressure. Although they are not always the protagonists, they remain competitive, a trait reflected in both the odds and their tactical organization. This group includes mid-table teams with solid performance, as well as some that had more consistent campaigns. The presence of Palmeiras and São Paulo, both with experienced squads and a history of defensive solidity, reinforces the profile of this cluster.

Based on the data from Figure 4(e), Cluster 5 comprises highly offensive teams, characterized by intense attacking volume, sustained pressure in the opponent's half, and a high frequency of set-piece opportunities. These teams are consistently favored in betting odds due to their aggressive positioning and relentless goal-seeking approach, though this style also leads to tactical vulnerabilities. Notably, Grêmio and Santos epitomize this profile, combining rapid transitions, sustained attacking pressure, and a high shot conversion rate.

Figure 4(f) demonstrates that Cluster 6 groups teams with striking characteristics related to offensive inefficiency and a more defensive playing style, typically attempting many shots during matches but with little accuracy. This is evidenced by the strong negative impact associated with the total number of off-target shots and the highest number of missed shots in a single match (Dwyer and Young, 2024). This combination indicates that while creating opportunities, these teams have clear difficulties in finishing plays. Teams tend to carry some favoritism in certain games, as shown by the influence of average odds, though this favoritism doesn't always translate into performance, especially when the team in question, like Corinthians, is in poor form. Lower odds (indicating favoritism) have a positive impact, while higher odds (indicating less market confidence) are associated with worse performances, which fits well with Chapecoense, generally seen as underdogs. Another key point is that the high number of fouls committed by teams in this cluster often reflects a more physical or reactive approach that may compromise game rhythm. The prediction suggests these teams perform slightly better when adopting a more restrained offensive posture, perhaps opting for a more conservative strategy. The presence of older players is also notable, particularly when observing the maximum age value of athletes, which tends to negatively affect predicted performance, likely by limiting

game intensity, physical recovery capacity, and transition speed.

4.4 Team Placement Dispersion and Cluster During the Championship

To better understand the behavior of the teams over time, Figure 5 shows the league standings for each cluster.

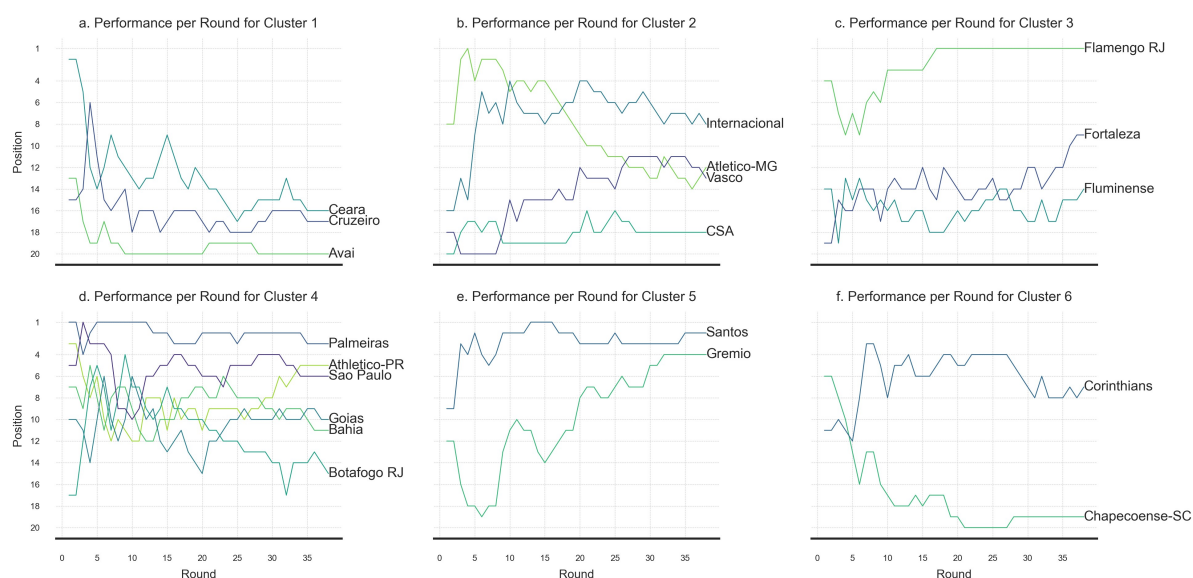


Figure 5: Graph Team placement dispersion and cluster during the championship.

Cluster 1, shown in Figure 5(a), is characterized by a constant battle against relegation. This group, composed of Avaí, Ceará, and Cruzeiro, displays the championship's worst average performance. From the very first rounds, these clubs occupied the bottom positions and remained near the relegation zone throughout the competition. The Cluster 1 line in the first graph stays consistently at the bottom of the table after the conclusion of the first round, indicating these teams were technically weaker with little ability to recover during the tournament.

Cluster 2, shown in Figure 5(b), is characterized by a promising but unstable start. Comprising Atlético-MG, CSA, Internacional, and Vasco, this group showed significant fluctuation, particularly in the early rounds. While some clubs started reasonably well, their performance throughout the championship was marked by inconsistency. The cluster's average trajectory suggests either a performance decline or difficulty sustaining results, remaining predominantly in the bottom half of the table.

As shown in Figure 5(c), Cluster 3, composed of Flamengo RJ, Fluminense, and Fortaleza, presents a distinct performance profile. Although the group as a whole falls within a mid-table range in the round-by-round standings graph, this average is skewed by Flamengo's strong performance as the competition champions. Thus, Cluster 3 can be interpreted as a group with consistent performances, featuring one outlier (Flamengo) that raised the group's standard while not substantially altering the average trajectory of the collective performance.

Figure 5(d) presents Cluster 4, composed of Atlético-PR, Bahia, Botafogo RJ, Goiás, Palmeiras, and São Paulo. This is the largest group and represents competitive teams that challenged for top-six positions (G6) without necessarily excelling. The cluster's trend line in the first graph occupies the upper-mid table range, indicating solid average performance but lacking the consistency required to seriously compete for top positions.

Cluster 5, shown in Figure 5(e), is characterized by having the highest average performance. Composed of Grêmio and Santos, its trend line remained in top positions throughout virtually the entire tournament, with both teams demonstrating remarkable consistency and competitiveness. The stability and sustained performance of these clubs established them as strong title contenders.

Cluster 6, as evidenced in Figure 5(f), represents unfulfilled expectations. Composed of Chapecoense-SC and Corinthians, this cluster stood out for displaying mediocre to poor performance, with consistent positioning in the lower half of the table. The most intriguing aspect here is Corinthians' presence alongside Chapecoense, indicating that, despite the historical differences between these clubs, their performances in that particular tournament were statistically similar in terms of volatility and table positioning. The cluster's trend line remains in the lower-mid table range, revealing underwhelming performance relative to expectations.

5 Conclusion

This study primarily aimed to explore the use of machine learning techniques, with emphasis on combining data clustering methods, a Random Forest classifier, and interpretative SHAP analysis, to delineate performance profiles of teams in the Brazilian Soccer Championship.

The integration of these approaches enabled a structured and explainable understanding of the patterns characterizing different collective team behaviors throughout the competition.

The results demonstrated that applying these techniques in sports contexts can not only enhance the analytical capabilities of professionals in the field but also strengthen data-driven decision-making. The cluster analysis revealed six distinct team profiles, ranging from groups with aging squads and low competitiveness to intensely offensive teams with high shot volumes and constant presence in the opponent's half. The study also highlighted groups featuring young, dynamic squads with solid and disciplined performances, as well as clusters of teams with low offensive efficiency despite high shot counts, employing more physical or reactive strategies. This segmentation underscores the championship's tactical and structural diversity, providing a concrete foundation for predictive analysis and more informed strategic decisions.

The Random Forest implementation identified key determining variables, including: goal totals, starting lineup's average age, win percentage against higher-ranked opponents, betting market odds, and foul counts, all suggesting direct impacts on collective performance. The correlation between these characteristics revealed a novel perspective for performance analysis in soccer analytics.

The limitations of this study may be found in the sample size (i.e., the number of teams considered from only one specific championship) and in the variables used, as several other factors could characterize teams' playing styles, such as ball possession, for example. Additionally, teams' historical performance in other competitions or derby victories could also influence results from a psychological perspective.

For future studies, we recommend enhancing the proposed model by incorporating additional statistical and contextual variables, such as ball possession, pass completion rates, average player ratings per match, weather conditions, stadium attendance figures, data from concurrent tournaments, player tracking (GPS) data, and even physiological metrics. These additions could significantly enrich the analyses and enhance their practical applicability in real-world sports management scenarios. Furthermore, integrating hybrid approaches that combine clustering with supervised predictive models could expand the methodology's potential for match outcome prediction and tactical strategy optimization.

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